Meet Rajeshkumar Dadhania

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.45
R. D. Contractor High School	Class XII (Gujarat Secondary and Higher Secondary Education Board)	2020	79.69%
Sincerite English School, Bardoli	Class X (Gujarat Secondary and Higher Secondary Education Board)	2018	91.00%

RESEARCH PROJECTS

Benchmarking Adaptability of Hardware Offloaded Congestion Control Algorithms [paper]

- Developed a programmable benchmarking tool for hardware-offloaded CCAs and emerging transports (**UET, Falcon, MRC**).
- Achieving **20× higher flexibility** to emulate realistic AI workloads at 100+ Gbps with **0.1 µs precision** versus existing works.
- Bridged hardware fidelity with software flexibility through prog. congestion-signal injection, enabling accurate measurement.
- Profiled RNIC congestion control across **ECN, INT, CSIG, and trimming**, exposing reaction to dynamic network conditions.
- *Stack: RDMA, RoCE v2, RNIC, DPU, Congestion Control, UET, Falcon, NCCL, Tofino/P4, DPDK, eBPF, XDP, Open vSwitch*

Simulator for Heterogeneous LLM Training [paper]

- **Developed** a heterogeneous-aware simulator for distributed LLM training across mixed GPU clusters and network speed.
- **Designed** abstractions for non-uniform workload partitioning and hybrid **DP+TP+PP** execution across heterogeneous devices.
- Implemented unified tensor resharding and collectives for mismatched parallelism, **reducing 21%** synchronization overhead.
- **Validated** predicting training time against real-world deployments with **<5% error** across most heterogeneous configurations.
- *Stack: C++, ns3, SimAI, NCCL, RDMA, Rail-optimized/Rail-only, Fat-tree topologies*

PUBLICATIONS

"BEACON: Benchmarking Adaptability of Hardware-Offloaded Congestion Control Algorithms" - ACM SIGCOMM NAIC,

2026

"Simulating LLM Training Workloads for Heterogeneous Compute and Network Infrastructure" - ACM SIGCOMM NAIC, 2025

"Simulating Unified Tensor Resharding in Heterogeneous AI Systems" - arXiv preprint, 2026 (EuroSys 2027 under review)

PROJECTS

ApplyMate -- AI job application assistant [github] [Chrome Extension]

- Built an AI agent that automates job application form filling across sites using Gemini API with human in the loop conformation.
- Automated complex forms filling with cross-site answer memory, validated by a 42-check Playwright end-to-end testing suite.

Hiring Management System [report]

- **Streamlining** hiring workflow with a full-stack hiring platform managing applications through **swipe-able decisions**.
- **Designed** relational schemas and REST APIs with server-side validation to enforce consistent candidate stage transitions.

Curriculum Management System [github]

- **Developed** a university-wide curriculum platform enabling professors to manage and publish subject syllabi across semesters.
- **Automated** versioned syllabus management and consolidated semester-wise PDF generation for academic year and branch.

EXPERIENCE

Software Engineer Intern

Zeus Learning, Mumbai

- Migrated a production LMS from .NET 5 to .NET 7, resolving API breaks and dependency conflicts, improving maintainability.
- Developed a full-stack hiring management platform streamlining application intake through swipe-able hiring decision.

SKILLS

Programming: C, C++, C#, Python, Java, JavaScript, .NET, Spring Boot, React, Angular, OOPs, Data Structures & Algorithm

Systems: Linux/Unix, Bash scripting, Kernel modules, Drivers, ARM SoC, Virtual memory, NUMA, Hugepages, PCIe, Git

Networks & Protocols: TCP/IP, UDP, RDMA, RoCE, InfiniBand, L2/L3 forwarding, Ethernet, Congestion Control, UET, MRC

Hardware: DPU, RNIC, SmartNIC, Marvell OCTEON, Data Accelerator Offload (DAO), Nvidia ConnectX, DOCA, P4, Tofino, TNA, BFRT, DPDK, eBPF, XDP, Open vSwitch, SR-IOV, DMA, RDMA, Zero-copy I/O

Distributed AI Training: NCCL, Pytorch, Parallelism, Tensor resharding, Simulation, Fat-tree, rail only/optimized topologies

Research: Systems research, Technical writing, Solution architect, Planning, Team leadership, Evaluation design, Data analysis

POSITIONS OF RESPONSIBILITY

Project Lead

NetX Lab, IIT Hyderabad

- **Leading** a research team, driving technical direction, task allocation, and delivery of the ACM SIGCOMM NAIC 2026 paper.
- **Deployed** hardware set-up and testbed integration with programmable switch and RNICs enabling end-to-end evaluation.

Teaching Assistant

- Assisted courses: **Computer Networks, Topics in Networks,** and **Distributed AI Training** through labs and tutorials.
- **Mentoring** students by guiding projects, evaluating assignments, and clarifying systems and networking concepts.

SCHOLASTIC ACHIEVEMENTS

- Achieved **3rd/109 position** in Foundations of Machine Learning Course Hackathon.

RELEVANT COURSES

Advanced Computer Networks | Foundations of Machine Learning | Deep Learning | Representation Learning | Hardware Architecture for Deep Learning | Topics in Network | Formal Verification Methods for AI | Advanced Data Structure & Algorithms

EXTRACURRICULAR

- Organising Volunteer -- **ACM ARCS 2026**, IIT Hyderabad
- Organising Volunteer -- **ICDCN 2025** (26th International Conference on Distributed Computing and Networking)
- Organising Volunteer -- **EBSB club**, IIT Hyderabad